

# DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

## Supplementary Winter Examination - 2023

Course: Third Year B. Tech. (Sem VI)

Branch: Civil Engineering

Subject Name: Water Power Engineering

Subject Code: BTCVPE604D

Max Marks: 60

Duration: 3 Hrs.

Date: 23-01-2024

### Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

### Q. 1 Solve Any Two of the following.

- |                                                                                 |      |   |
|---------------------------------------------------------------------------------|------|---|
| A) Enlist types of hydropower plant; explain any one in brief.                  | CO 1 | 6 |
| B) Write a detail note on Components of Water Project.                          | CO 1 | 6 |
| C) Explain following terms: 1) Load Factor 2) Load curve 3) Load Duration Curve | CO2  | 6 |

### Q.2 Solve Any Two of the following.

- |                                                                                                          |          |   |
|----------------------------------------------------------------------------------------------------------|----------|---|
| A) Define intake, give its type and explain any one.                                                     | Remember | 6 |
| B) What are anchor blocks and explain any 4 ways in which they might be used in water power engineering. | CO2      | 6 |
| C) Explain basic design criteria of simple surge tank.                                                   | CO 1     | 6 |

### Q. 3 Solve Any Two of the following.

- |                                                                                                  |          |   |
|--------------------------------------------------------------------------------------------------|----------|---|
| A) Sketch the typical arrangement of a hydro power station and explain its different components. | Remember | 6 |
| B) Give types of underground power house. Explain any one in detail.                             | CO2      | 6 |
| C) What are the advantages and disadvantage of underground power house?                          | CO2      | 6 |

### Q.4 Solve Any Two of the following.

- |                                                                |      |   |
|----------------------------------------------------------------|------|---|
| A) Explain with neat sketch Kaplan turbine and Pelton turbine. | CO 1 | 6 |
| B) Explain the ways to avoid cavitation.                       | CO 1 | 6 |
| C) Explain draft tube & give its types.                        | CO 1 | 6 |

### Q. 5 Solve Any Two of the following.

- |                                                                                     |          |   |
|-------------------------------------------------------------------------------------|----------|---|
| A) Enlist the components of pumped storage plants and explain them.                 | Remember | 6 |
| B) Explain tidal power stations. Write down the advantages of tidal power stations. | CO 2     | 6 |
| C) Write a note on economics of pumped storage plants.                              | CO 3     | 6 |

\*\*\* End \*\*\*